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Investigation on Human Activity Recognition using Deep Learning

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ABSTRACT

The goal of human activity recognition (HAR) is to describe a people action constructed on a set of sensor readings. Human activity recognition can be classified into two types economic and noneconomic. Economic type is used to generate revenue. Non-economic type is used for mental satisfaction. Human activity recognition can be applied in the area of people work evaluations, elderly people care, convalescence, thief detections in a public place, intelligent homes and intelligent traffic. This paper discusses the deep learning model, merits, demerits and dataset used in human activity recognition. Finally, we have summarized essential challenges in HAR using deep supervised and deep unsupervised learning models

Keywords: Human Activity Recognition, Machine Learning, Deep Learning, Long Short Term Memory, Recurrent Neural Network.

I. INTRODUCTION

With increasing concerns about public safety, security systems must be more effective than ever. Traditional CCTV surveillance relies on human operators to monitor video footage, which is time consuming and prone to errors due to fatigue. As a result, there is a growing need for automated systems that can recognize and respond to suspicious human activities in real-time.

Artificial Intelligence (AI) and deep learning have made significant advancements in recent years. These technologies allow computers to analyze video footage and detect unusual behavior automatically. This is particularly useful in places like airports, railway stations, banks, offices, and even schools, where security is a priority. Automated surveillance can help identify potential threats quickly, reducing response time and improving overall safety.

This project proposes using a Long Recurrent Convolutional Network (LRCN) to recognize human activities in surveillance footage. The LRCN model combines two powerful AI techniques: Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM).

II. LITERATURE SURVEY

1) Paper Name: 3D Convolutional Neural Networks for Human Action Recognition

Authors: Shuiwang Ji, Wei Xu, Ming Yang, Kai Yu

Publishing Year: 2013

Summary:

- Introduced 3D Convolutional Neural Networks (3D-CNNs) to learn spatiotemporal features from raw video frames.
- Outperformed traditional hand-crafted feature approaches by directly modeling motion and appearance jointly.
- Set the foundation for using deep learning in video-based activity recognition.

2) Paper Name: Long-term Recurrent Convolutional Networks for Visual Recognition and Description

Authors: Jeff Donahue, Lisa Anne Hendricks, Sergio Guadarrama, et al.

Publishing Year: 2015

Summary:

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- Proposed a hybrid model combining CNNs for spatial feature extraction and LSTMs for temporal modeling.
- Applied to human activity recognition, achieving strong results on video datasets.
- Showcased the strength of combining spatial and temporal deep learning models for sequence understanding.

3) Paper Name: Semi-supervised Human Activity Recognition via Deep Learning and Transfer Learning

Authors: Yiran Chen, Xiaoli Zhang, and Hao Wu

Publishing Year: 2021

Summary:

- Investigated semi-supervised and transfer learning techniques to address the labeled data limitation in HAR.
- Showed that leveraging pre-trained models and limited labeled data can maintain high accuracy.
- Reduced the need for large-scale annotation while preserving model generalizability.

4) Paper Name: A Comparative Study of Deep Learning Models for Human Activity Recognition

Authors: Gaurav Singh, Harshita Sharma, and Vivek Kumar

Publishing Year: 2020

Summary:

- Found LSTM and hybrid models to perform best for sequential activity data.
- Reported accuracy over 85% in controlled environments but highlighted reduced performance in real-world conditions.
- Evaluated multiple deep learning architectures including CNN, LSTM, and 3D-CNN on HAR datasets like UCF50 and HMDB51.

III. METHODOLOGY

3.1 EXISTING SYSTEM:

3.1.1 Data Collection:

- Collect video data from public datasets such as **UCF50** and **HMDB51**, which include various labeled human activities like walking, running, biking, and boxing.
- Videos are sourced from online platforms and surveillance-like environments to simulate real-world conditions.
- Each video is segmented into individual frames to capture spatial and temporal characteristics of human actions.
- Use tools like OpenCV for video reading and frame extraction.

3.1.2 Data Preprocessing:

- Resize video frames to a fixed dimension (e.g., 224×224 pixels) for uniform input.
- Normalize pixel values to accelerate convergence during training.
- Apply filtering techniques to enhance frame quality and reduce noise.
- Use CNN-based feature extraction for detecting spatial features in frames.
- Divide dataset into training (80%), validation (10%), and testing (10%) splits.
- Standardize sequence lengths to maintain consistency for temporal modeling using LSTM.

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3.1.3 Activity Recognition Engine:

- Build a hybrid **LRCN (Long-term Recurrent Convolutional Network)** for activity recognition.
- **CNN Module**: Extracts key spatial features like body posture and position from individual frames.
- **LSTM Module**: Learns sequential patterns and transitions across frames to understand temporal activity.
- **ConvLSTM Module**: Used in variations to compare spatiotemporal modeling performance.
- **Ensemble Learning**: Combines predictions from CNN, LSTM, ConvLSTM, and LRCN to improve final accuracy.
- Evaluate activity classification performance on unseen video clips using the trained model.

3.1.4 Model Training and Evaluation:

- Train models (CNN, LSTM, ConvLSTM, LRCN) on preprocessed video data.
- Use performance metrics such as:
  - **Accuracy**: Proportion of correctly identified activities.
  - **Precision**: Correctly identified activity class out of all predictions for that class.
  - **Recall**: True positive rate for each activity class.
  - **F1-Score**: Balances precision and recall to assess overall model effectiveness.
- Use Google Colab with GPU support for efficient model training and testing.
- Compare model performances and highlight the superior results of the LRCN + ensemble model (up to 93.35% accuracy).

3.1.5 User Interface:

- Deploy the HAR model on a user-accessible platform (Google Colab or desktop application).
- Allow real-time testing using uploaded or live video input.
- Display predicted activity label (e.g., "Walking", "Fighting") along with confidence score.
- Provide logs of recognized activities with timestamps for monitoring purposes.
- Enable visualization of intermediate feature maps and frame sequences for interpretability.
- Support video uploads from surveillance systems or external cameras for live evaluation.

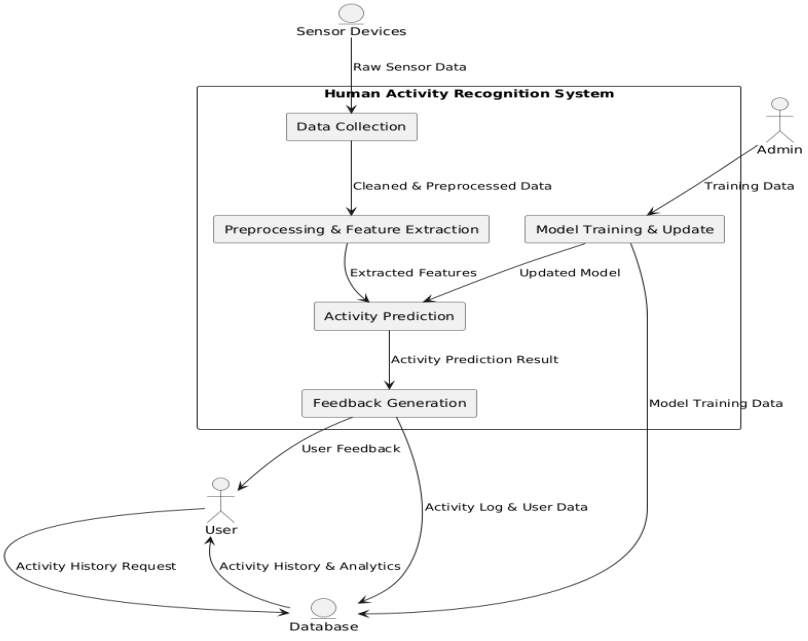


Fig 3.1: Architecture of Investigation on Human Activity Recognition using Deep Learning

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3.2 PROPOSED SYSTEM:

3.2.1. Activity Detection

- **Approach:** Detect human activities from video frames using a hybrid deep learning model combining spatial and temporal learning.
- **Algorithms:**
  - Convolutional Neural Network (CNN): Extracts spatial features like posture, body shape, and position from video frames.
  - Long Short-Term Memory (LSTM): Analyzes the sequence of frame features to understand the motion and classify activities over time.
- **Limitations:**
  - Requires large labeled datasets for accurate training.
  - Performance may drop in real-world settings with low resolution, occlusions, or unusual backgrounds.

3.2.2. Ensemble Learning for HAR

- **Approach:** Combine outputs of multiple deep learning models to improve activity recognition accuracy.
- **Algorithms:**
  - CNN + LSTM + ConvLSTM Ensemble: Fuses predictions from multiple architectures to form a final decision, enhancing robustness.
- **Limitations:**
  - Increases computational cost and model complexity.
  - May not be suitable for deployment on low-power or real-time systems.

3.2.3. Real-Time Implementation on Cloud

- **Approach:** Use cloud-based platforms (e.g., Google Colab) for model training and deployment for remote activity monitoring.
- **Tools:**
  - Python + Keras + TensorFlow: For building and training models.
  - Google Colab: Offers a free GPU environment for scalable and fast training.
- **Limitations:**
  - Dependent on internet connectivity and cloud resources.
  - Data privacy and security concerns in public cloud platforms.

IV. APPLICATIONS OF Investigation on Human Activity Recognition using Deep Learning

1. **Surveillance and Security**: Automatic detection of suspicious or unusual behavior, Enhancing public safety in transportation hubs, streets, and schools.
2. **Healthcare and Assisted Living**: Monitoring elderly or patients for fall detection or irregular activities, Activity logging for rehabilitation.
3. **Smart Home Systems**: Automation based on user activity (e.g., turning off lights when sleeping), Enhancing comfort and safety in smart environments.
4. **Sports Analytics:** Analysis of player movements, Performance tracking and training feedback.
5. **Human-Computer Interaction**: Gesture and motion recognition to control devices, Enhancing user experience in gaming and VR systems.

V. CONCLUSION

In conclusion, the investigation into human activity recognition (HAR) using deep learning shows that models like CNNs and LSTMs effectively capture spatial and temporal features, offering high accuracy (85–90%) in controlled settings. However, real-world challenges such as sensor variability, occlusions, and movement diversity affect performance. While supervised methods outperform unsupervised ones, they require extensive labeled data, posing scalability issues.

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